

OPTIMAL INTRANSITIVE DICE SETS FOR REGULAR TOURNAMENTS.

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ABSTRACT. This note explores optimal non-transitive random variable tournaments, focusing on the case of 5 and 7 random variables.

1. INTRODUCTION

Consider a set of independent random variables X_1, X_2, X_3 , such that the probability of each variable being larger than the next is more than 50%, that is $P(X_1 > X_2) > 0.5$ and $P(X_2 > X_3) > 0.5$. Trybuła [1] demonstrated a paradox that random variables need not exhibit transitivity in terms of probability of one value being larger than another, meaning it is possible that $P(X_1 > X_3) < 0.5$. One natural way to conceptualize random variables is dice. Thus, a classical example of this intransitive dice is Efron's dice, a set of four regular 6-sided dice $\{D_1, D_2, D_3, D_4\}$ with the following configurations of values on the faces:

Faces	D_1	D_2	D_3	D_4
1 and 2	4	3	6	5
3	4	3	2	5
4	4	3	2	1
5 and 6	0	3	2	1

In this example, $P(D_1 > D_2) = P(D_2 > D_3) = P(D_3 > D_4) = P(D_4 > D_1) = 2/3$. As it is shown independantly by Trybuła [2] and Usiskin [3], the maximal probability exhibited by a cycle of four intransitive random variables is exactly $2/3$, rendering Efron's dice optimal in this sense. The optimal probabilities for cycles of other numbers of intransitive random variables are irrational. For three variables, the optimal probability is the golden ratio conjugate $\phi^{-1} = \frac{\sqrt{5}-1}{2} \approx 0.618$, the positive root of $x^2 + x - 1$. Using the connection between the golden ratio and the Fibonacci numbers, it is possible to construct regular dice sets that approximate this optimal probability very well. This result was discussed in [4]. Let $F_1 = F_2 = 1$ and $F_{n+2} = F_{n+1} + F_n$ and consider the set of three F_{2n} -sided dice (for $n > 2$):

Faces	D_1	D_2	D_3
$1 \dots F_{2n-1} - 1$	2	8	5
F_{2n-1}	4	6	3
$F_{2n-1} + 1 \dots F_{2n} - 1$	9	1	5
F_{2n}	9	1	7

Considering Cassini's identity $F_{2n-2}F_{2n} + 1 = F_{2n-1}^2$, we observe

$$P(D_1 > D_2) = P(D_2 > D_3) = P(D_3 > D_1) = \frac{F_{2n-1}F_{2n} - 1}{F_{2n}^2} \xrightarrow{n \rightarrow \infty} \phi^{-1}.$$

Amusingly, this can be emulated in the real world quite nicely with pairs of dodecahedra with $12^2 = 144$ outcomes each, which is luckily the twelfth Fibonacci number, so the winning probability is $\frac{89 \cdot 144 - 1}{144^2} = 0.618007$ only different from ϕ^{-1} beyond the fifth digit. In the continuous limit, we can consider a set of three 'golden' unfair coins $\{C_1, C_2, C_3\}$, each having $P(\text{Heads}) = \phi^{-1}$ and $P(\text{Tails}) = 1 - \phi^{-1} = \phi^{-2}$:

Side	C_1	C_2	C_3
Heads	2	4	3
Tails	5	1	3

2. REGULAR INTRANSITIVE TOURNAMENT WITH 5 VARIABLES

2.1. Fair dice (each side has the same probability). For a larger number of random variables, it is also interesting what the probabilities are for all pairs of dice in a complete graph. Akin and Saccamano [5] showed the construction of an arbitrary tournament with a given number of dice and faces on each die. In particular, arbitrary tournament of $2n + 1$ dice can be constructed with 3-sided dice and numbers $\{1 \dots 3(2n + 1)\}$ on the faces. This means that each probability $P(D_i > D_j) = 5/9$. But it can be higher. For example, with 5-sided dice we can construct the set

Side	D_1	D_2	D_3	D_4	D_5
1	2	3	1	7	6
2	5	4	10	12	8
3	11	16	14	13	9
4	23	20	19	15	17
5	24	22	21	18	25

where $P(D_i > D_j) = 0.56 > 5/9$ for $j - i \equiv 1$ or $2 \pmod{5}$. It can be emulated with regular icosahedra, where each value is mapped to four faces. The natural question is how high can the winning probability be in a regular tournament.¹

2.2. A general case of irrational unfair dice. Let's consider intransitive systems of with dice of uneven, irrational probabilities. Such loaded dice would be difficult to emulate physically, and the most feasible way to implement irrational probabilities are spinning wheels, not dice, but I will use the word dice throughout, as a synonym of a random value. Surprisingly, a simple degree-3 polynomial $x^3 + x^2 - 1$ can generate a decent intransitive tournament with uniform dominance probability of almost 57.0%. Consider the positive

¹I also provide set of 11-sided dice tournament with more than 57% winning probabilities on my website <https://belliavesha.github.io/second.html#dice>

real root ψ of $x^3 - x - 1 = 0$, so

$$\begin{aligned}\psi &\approx 0.75487766624669276005, \\ \psi^2 &\approx 0.56984029099805326591, \\ \psi^3 &\approx 0.43015970900194673409, \\ \psi^4 &\approx 0.32471795724474602596, \\ \psi^5 &\approx 0.24512233375330723995,\end{aligned}$$

and note that $1 = \psi^2 + \psi^3 = \psi^3 + \psi^4 + \psi^5 = \psi + \psi^5$.

The tournament is realized as 3-faced dice with the following values and probabilities:

Side prob.	D_1	D_2	D_3	D_4	D_5
ψ^3	5	7	9	2	1
ψ^4	5	4	3	6	8
ψ^5	5	4	3	11	10

For each pair of dice

$$P(D_i > D_j) = \psi^2 \quad j - i \equiv 1, \text{ or } 2 \pmod{5}.$$

This set can be generalized for the case of $2n + 1$ dice as follows:

Side prob.	D_1	D_{1+k}	D_{n+1+k}
ψ^3	$2n + 1$	$2n + 1 + 2k$	$n + 1 - k$
ψ^4	$2n + 1$	$2n + 1 - k$	$2n + 2k$
ψ^5	$2n + 1$	$2n + 1 - k$	$5n + 2 - k$

defined for $k = 1, \dots, n$. Here each pair of dice

$$P(D_i > D_j) = \psi^2 \quad j - i \equiv 1, \dots, n \pmod{2n + 1}.$$

Alternatively, this can be achieved with a similar system:

Side prob.	D_1	D_{1+k}	D_{n+1+k}
ψ^3	$2n + 1$	$2k$	$3n + 2 - k$
ψ^4	$2n + 1$	$4n + 2 - k$	$2k - 1$
ψ^5	$4n + 2$	$4n + 2 - k$	$5n + 3 - k$

It seems that the maximum common probability for a 5-tournament can be obtained by generalizing the case of the 'golden' coin. In fact, let ψ be the positive real root of $x^4 + x - 1$, and note that $\psi^3 + \psi^4 + \psi^5 + \psi^6 = 1$. Then let us consider a set of 5 four-sided dice with corresponding probabilities for each side $\{\psi^3, \psi^4, \psi^5, \psi^6\}$:

Side prob.	D_1	D_2	D_3	D_4	D_5
ψ^3	19	18	17	14	8
ψ^4	3	2	1	5	9
ψ^5	4	6	10	12	20
ψ^6	11	15	16	13	7

For this set,

$$P(D_i > D_j) = \frac{1}{1 + \psi} \approx 0.57988 \quad j - i \equiv 1 \text{ or } 2 \pmod{5}$$

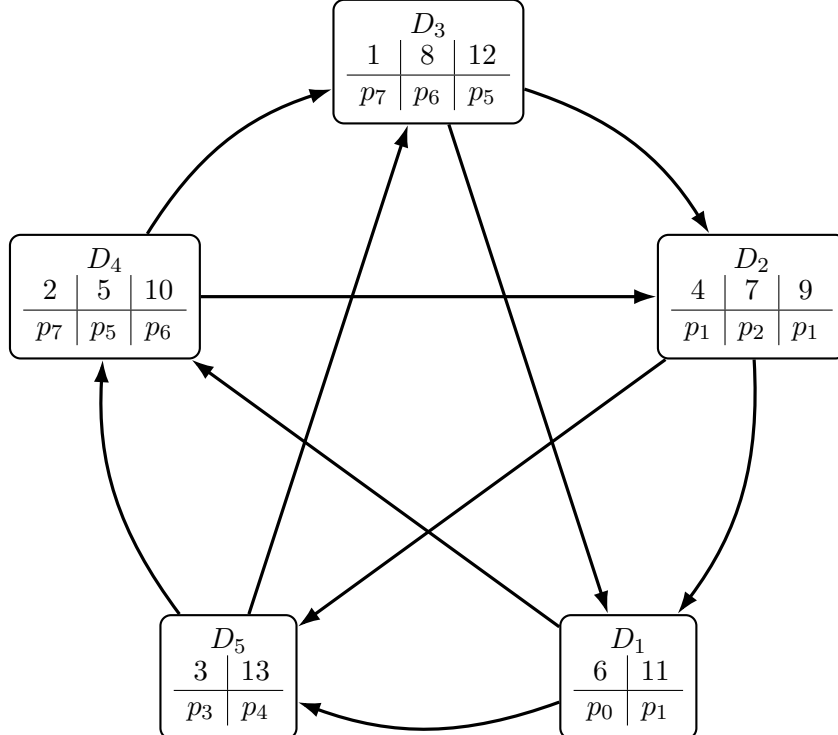
However, it is not straightforward to generate regular n -sided dice set that approximates this probability as it was with Fibonacci numbers, because the (1, 4)-bonacci numbers (such that $f_{n+1} = f_n + f_{n-3}$) do not seem to have enough identities similar to Cassini's.

Surprisingly, this is not quite the maximum for a 5-tournament. If we relax the condition of all 5 dice being the same side probabilities, we can find a nice set for 5-tournament with probability $> 58\%$.

Define the first side probability p_0 as the positive real root of the polynomial $2x^4 - x^3 + x - 1$. The approximate value is $p_0 \approx 0.764192$. The next face probabilities can be constructed using the following auxiliary constants derived from p_0 :

$$\begin{aligned} p_1 &= 1 - p_0 \approx 0.2358 \\ p_2 &= 2p_0 - 1 \approx 0.5284 \\ p_3 &= p_0^2 \approx 0.5840 \\ p_4 &= 1 - p_3 \approx 0.4160 \\ p_5 &= 2p_3 - 1 \approx 0.1680 \\ p_6 &= 1/p_0 - p_0 \approx 0.5444 \\ p_7 &= p_2p_6 \approx 0.2876 \end{aligned}$$

And it can be shown that $p_7 = p_5/p_3 = 1 - p_5 - p_6$. The diagram below illustrates the five dice. Each node represents a die with its face values and their associated probabilities. The arrows indicate the direction of dominance (e.g., $D_2 \rightarrow D_3$ means $P(D_4 > D_0) = p_3$). The value $p_3 \approx 58.4\%$ is the common winning probability in the tournament.



The hypothesis is that the aforementioned sets of random values are optimal, that is, achieving the maximum winning probability in tournaments of size 5.

3. THE CASE WITH 7 VARIABLES AND BEYOND

3.1. The 'golden' seven. Let's consider another seven-variable intransitive system, using the polynomial $x^6 + x - 1$. For tournaments of size $2n+1 > 7$, similar 'golden' constructions will not yield optimal variables, because for each positive real root ψ_{2n+1} of $x^{2n} + x - 1$, the value

$$\frac{1}{1 + \psi_{2n+1}} < \frac{5}{9}.$$

That leaves the last number of dice, 7, for which this pattern would make sense to try:

$$\frac{1}{1 + \psi_7} = 0.5624 \dots > \frac{5}{9},$$

where $\psi_7 = 0.7780895986786 \dots$, the root of $x^6 + x - 1$.

Such a set is given in the following table:

Side prob.	D_1	D_2	D_3	D_4	D_5	D_6	D_7
ψ_7^5	17	14	11	8	27	31	20
ψ_7^6	24	28	32	38	3	2	37
ψ_7^7	18	15	12	9	41	22	1
ψ_7^8	25	29	33	39	6	40	21
ψ_7^9	19	16	13	10	42	5	35
ψ_7^{10}	26	30	34	36	7	23	4

In this set

$$P(D_i > D_j) = \frac{1}{1 + \psi_7} \quad j - i \equiv 1, 2, \text{ or } 3 \pmod{7}$$

There are two more regular tournaments on 7 vertices, and, they are most probably also constructable with this scheme.

Seven-star tournament. The full tournament contains $\binom{7}{2} = 21$ directed edges. Figure 1 shows the compact star realization, with arrows pointing from the smaller to the larger variable.

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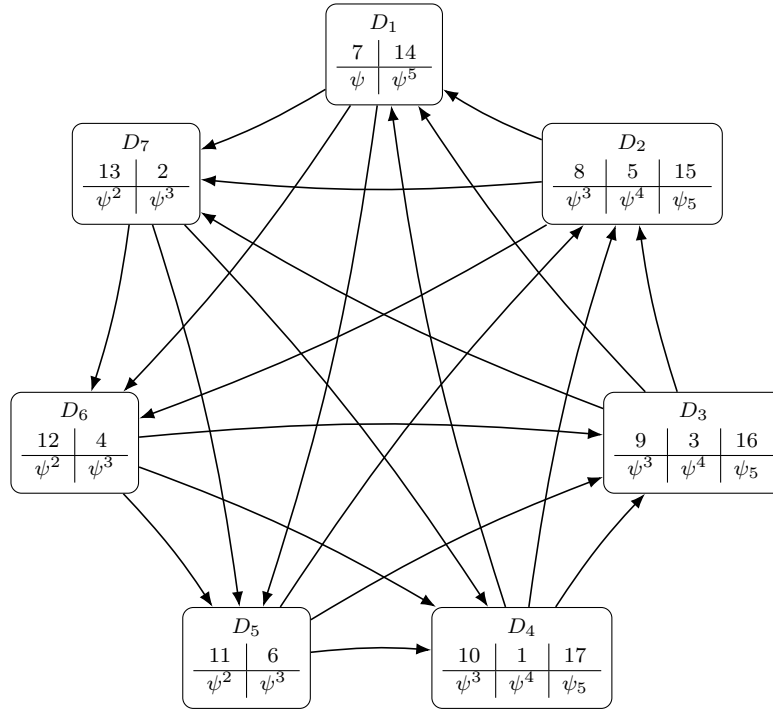


FIGURE 1. Seven-way intransitive dice tournament. Each node represents a die with face values and symbolic probabilities. Arrows indicate dominance with probability $\psi^2 \approx 0.56984$.

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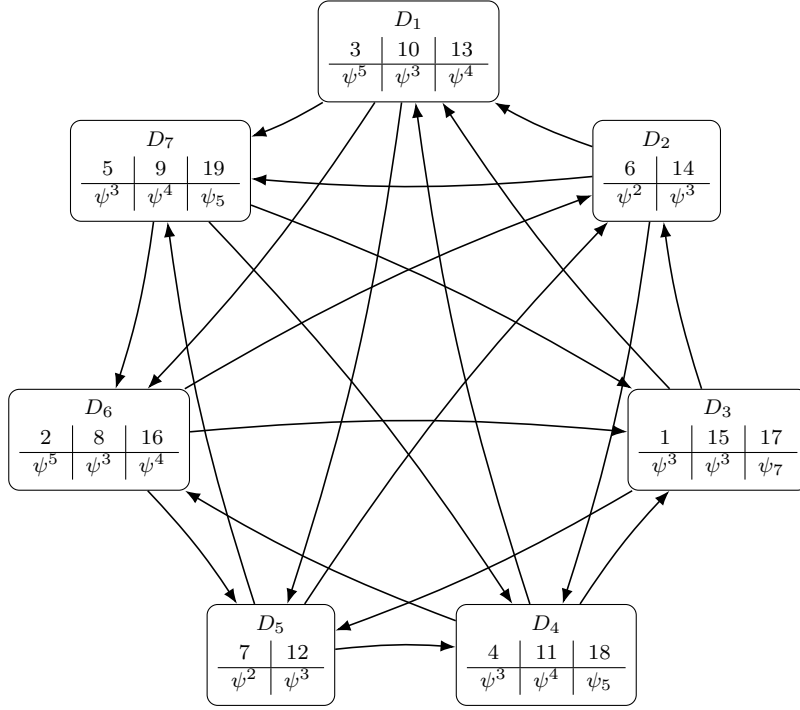


FIGURE 2. Seven-way intransitive dice tournament. Each node represents a die with face values and symbolic probabilities. Arrows indicate dominance with probability $\psi^2 \approx 0.56984$.